The Ring Fountain

SYPT Reporter Presentation

Nikola Kocovic

Team: Quantum Dots

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Problem Statement



Figure: The Exp Is Very Dangerous!

Ring Fountain

"When a flat metal **ring** falls from a certain height into a water tank, it generates a fountain that can shoot water high into the air. How does the **maximum height** of the fountain depend on the ring's parameters?"

System Parameters

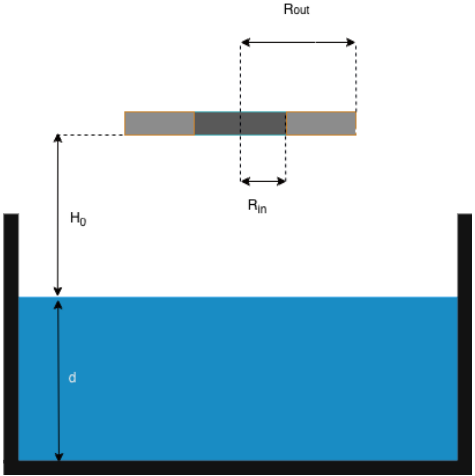


Figure: Primary Parameters

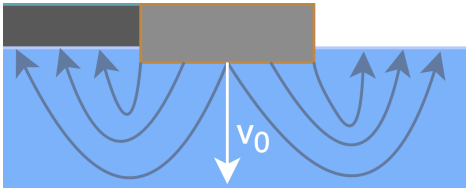
Primary Parameters

- **Inner Radius** R_{in}
- **Outer Radius** R_{out}
- **Initial Height** (H_0)
- **Water depth** (d)

Physical Constants

- Water Density (ρ)
- Surface Tension (σ)
- Gravitational Acceleration (g)

Qualitative explanation: Impact phase



- **Disc Velocity Persistence:** In the high Reynolds number regime ($Re = \frac{\rho v(R_{out}-R_{in})}{\mu} \approx 50000 \gg 1$), inertial forces dominate. The velocity of the disc remains effectively **constant** during its movement through the fluid volume.
- **Irrelevance of Gravity (g):** Pressure generated by the impact is several orders of magnitude higher than the hydrostatic pressure. **Gravity is neglected** for the fluid movement until the jet is launched.
- **Energy Transfer:** A pressure gradient in the fluid is created, tending to push the water from under the ring. That gradient supplies kinetic energy to the water in the jet.

Qualitative explanation: Motion of water

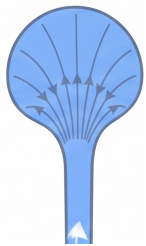


Figure: Qualitative Shape of the Jet for Higher Radii

- **Jet Creation:** The water is concentrated into the cavity in the ring, forming a region of a sharp pressure gradient. **The gradient is higher the smaller the inner radius is relative to the outer**. It pushes the water out through the cavity.
- **Flight of the Jet:** Here the water decelerates due to gravity, while focusing toward a sphere-like form (for higher radii).
- **Influence of surface tension:** Due to the **Rayleigh-Plateau instability** the jet will tend to break up into tiny droplets. Therefore **the thinner the jet is initially, the sooner it'll break into droplets**.

Relevant Dimensionless Numbers

Weber Number (We)

Characterizes the ratio of fluid inertia to surface tension, predicting the onset of the Rayleigh-Plateau instability:

$$We = \frac{\rho v_{jet}^2 R_{jet}}{\sigma} \tag{1}$$

($We \gg 50$ for minimisation of Rayleigh-Plateau instability)

Euler Number (Eu)

Characterizes the ratio of the impulsive pressure gradient to the inertial forces of the impact:

$$Eu = \Delta P / \rho v_0^2 \tag{2}$$

Is only $f(\beta)$ where $\beta \equiv \frac{R_{in}}{R_{out}}$

$$h \propto v_{jet}^2 \sim Eu \cdot v_0^2 \sim Eu \cdot H_0 \tag{3}$$

Qualitative Explanation: Importance of the Depth

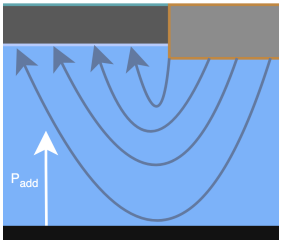


Figure: $d \sim R$

High Depth ($d \gg R$)

Water acts as an infinite bath

$d \sim R$

The bottom forces more water to accelerate into the jet, therefore the jet has more kinetic energy, so **it's higher**.

Shallow Water ($d < R$)

The water flow interacts with the bottom, with the impact of the disc with the bottom also interfering. The jet will be **significantly lower**.

Theoretical Framework: Assumptions

The fluid is treated as a **Perfect Fluid** justified by the following:

- **Incompressibility:** $\nabla \cdot \mathrm{u} = 0$. Fluid volume remains constant.
- **Inviscid Flow:** Viscous dissipation is negligible compared to inertial momentum transfer at the moment of jet formation.
- **Irrotationality:** Flow originates from rest through a potential impulse, meaning $\nabla \times \mathrm{u} = 0$ allowing the definition $\mathrm{u} = \nabla\phi$. Therefore **Laplace's equation $\nabla^2\phi = 0$ holds**.
- **Instanteneous energy transfer:** From slow motion footage we can see that the creation of the jet is almost instanteneous, which means, **for calculating the energy of the jet, we can consider only the impact phase**
- **Relevance of parameters:** Gravitational acceleration, and surface tension are irellevant during the impact phase.

Energy Storage in Potential Flow

Kinetic Energy Integral

The total kinetic energy of the fluid can be determined using Green's First Identity ($\nabla^2\phi = 0$):

$$E_{fluid} = \frac{1}{2}\rho \iiint_V |\nabla\phi|^2 dV = \frac{1}{2}\rho \iint_S \phi \frac{\partial\phi}{\partial n} dS \tag{4}$$

where S bounds the fluid volume and n is the outward normal.

The Surface Integral

The boundary integral vanishes on the tank walls and the tank floor, because $\frac{\partial\phi}{\partial n} = 0$ (the velocity is 0). It also vanishes on the free surface because the impulsive pressure is zero ($\phi = 0$).
The only non-zero contribution is exactly under the solid ring, where the normal velocity matches the impact speed ($\frac{\partial\phi}{\partial n} = v_0$):

$$E_{fluid} = \pi\rho v_0 \int_{R_{in}}^{R_{out}} \phi(r, 0)\, r\, dr \tag{5}$$

Theoretical Model: Laplace's Equation

Laplace's Equation

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r} + \frac{\partial^2\phi}{\partial z^2} = 0 \tag{6}$$

Separation of Variables

$\phi(r,z) = R(r)Z(z)$, the radial solution is governed by Bessel's equation, selecting the bounded Bessel function (sine wave in cyllindrical geometry) of the first kind $J_0(kr)$ (see Appendix 1), where k is the radial wavenumber. The vertical solution takes the form of hyperbolic functions.

Theoretical Model: The Confinement Effect

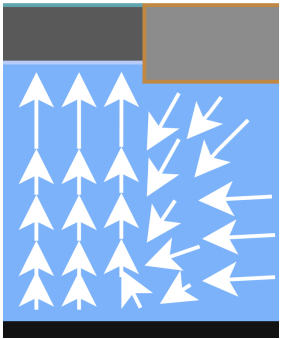


Figure: Confiniment Velocity Field

Kinematic Boundary Condition: Tank Floor

The solid tank floor is located at depth $z = -d$. The vertical velocity must vanish at this boundary:

$$\left.\frac{\partial \phi}{\partial z}\right|_{z=-d} = 0 \tag{7}$$

Effect on the Vertical Solution

Applying this boundary condition to the general vertical solution $Z(z)$ gives a solution of form:

$$Z(z) \propto \cosh\left(k(z+d)\right) \tag{8}$$

Theoretical Complexity: Mixed Boundary Conditions

Defining the Surface Partition ($z = 0$)

The boundary conditions are **mixed** and **discontinuous** at the impact surface:

$$\begin{cases} \phi(r,0) = 0 & r < R_{in} \\ \frac{\partial \phi}{\partial z}\big|_{z=0} = -v_0 & R_{in} \leq r \leq R_{out} \\ \phi(r,0) = 0 & r > R_{out} \end{cases} \tag{9}$$

The Mathematical Challenge

Standard separation of variables fails because the boundary type switches between **Dirichlet** and **Neumann** conditions across the radial domain. This necessitates the use of Integral Equations to solve for the spectral amplitude function $A(k)$.

Theoretical Model: Superposition of Wavenumbers

The General Potential Field

To satisfy the complex surface boundary conditions, we must sum over all possible wavenumbers k continuously from 0 to ∞, utilizing a Hankel transform. We introduce an amplitude function $A(k)$ and a normalizing $\sinh(kd)$ term for convenience at $z = 0$:

$$\phi(r, z) = \int_0^\infty A(k) J_0(kr) \frac{\cosh\left(k(z+d)\right)}{\sinh(kd)}\, dk \tag{10}$$

Surface Potential and Velocity

At the moment of impact ($z = 0$), the potential and vertical velocity reduce to:

$$\phi(r, 0) = \int_0^\infty A(k) \coth(kd) J_0(kr)\, dk \tag{11}$$

$$\left.\frac{\partial \phi}{\partial z}\right|_{z=0} = \int_0^\infty kA(k) J_0(kr)\, dk \tag{12}$$

Theoretical Model: Mixed Boundary Conditions

At the moment of impact ($t = 0$), the water surface ($z = 0$) is divided into three distinct zones, leading to a system of three integral equations to solve for $A(k)$:

- **Inner Free Surface ($0 \leq r < R_{in}$):** Zero impulsive pressure.

$$\phi = 0 \implies \int_0^\infty A(k) \coth(kd) J_0(kr)\, dk = 0 \tag{13}$$

- **Solid Impact Zone ($R_{in} \leq r \leq R_{out}$):** Velocity matches the ring.

$$\frac{\partial \phi}{\partial z} = -v_0 \implies \int_0^\infty kA(k) J_0(kr)\, dk = -v_0 \tag{14}$$

- **Outer Free Surface ($r > R_{out}$):** Zero impulsive pressure.

$$\phi = 0 \implies \int_0^\infty A(k) \coth(kd) J_0(kr)\, dk = 0 \tag{15}$$

Theoretical Model: Energy

Solving for Added Mass

The triple integral equations completely define the fluid state. While analytically intractable without approximations, solving them numerically yields $A(k)$, and thus the exact surface potential $\phi(r, 0)$.

Total Kinetic Energy

Using Green's First Identity, the total kinetic energy transferred to the fluid (E_{fluid}) can be evaluated by integrating the surface potential strictly over the area of the solid ring:

$$E_{fluid} = \pi \rho v_0 \int_{R_{in}}^{R_{out}} \phi(r, 0)\, r\, dr \equiv \frac{1}{2} M_a(R_{in}, R_{out}, d) v_0^2 = M_a(R_{in}, R_{out}, d) g H_0 \tag{16}$$

Partition Of Flow

Theoretical Decomposition

The potential flow generated by an impacting ring is topologically separated into two distinct regions by the ring's solid boundary:

1. **The Exterior Flow (Crown)**
   - Linked to the outer added mass M_{crown}.
   - Drives the outward radial splash.
   - Creates the visible "Crown" structure.

2. **The Interior Flow (Jet)**
   - Linked to the inner added mass M_{jet}.
   - Converges toward the central axis ($r \to 0$).
   - Propels the high-energy central jet.

Quantifying the Energy Partition

The total kinetic energy imparted to the water E_{total} is partitioned:

$$E_{total} = E_{crown} + E_{jet} \tag{17}$$

And since the jet focuses into a sphere,

$$E_{jet} \approx M_{jet} gh \tag{18}$$

Efficiency Estimates (due to Peters, Meer, Gordillo [3]) ($\beta \approx 0.2$ to $\beta \approx 0.7$)

- Calculates the proportion of energy that goes into the crown relative to the total energy transferred to the fluid.
- **Crown Energy Loss ($\approx 65\%$):** The majority of energy is consumed by the exterior flow due to the larger outer radius and displacement volume.
- **Jet Energy Proportion ($\approx 35\%$):** (The rest) A minority of the energy is directed inward, but it is focused into a significantly smaller volume, leading to high **energy density**.

Therefore,

$$M_{jet} gh \approx E_{jet} \approx \eta E_{total} = \eta M_a g H_0 \tag{19}$$

Theoretical Model: Final Height Equation

Effective Jet Mass ($d \sim R$)

For a shallow tank, the confinement limits the entrained mass to the fluid directly between the ring and the floor. Substituting $R_{in} = \beta R_{out}$, we extract the dimensionless depth d/R_{out}:

$$M_{jet} \approx \frac{\rho \pi d}{4}(R_{in} + R_{out})^2 = \frac{\rho \pi R_{out}^3}{4}\left(\frac{d}{R_{out}}\right)(1+\beta)^2 \tag{20}$$

Final Maximum Height Estimation for $d \sim R$

Equating the jet's potential energy to the partitioned kinetic energy ($M_{jet} g h_{max} \approx \eta M_a g H_0$) and isolating h_{max} yields:

$$h_{max} \approx \frac{8\eta\Gamma \int_\beta^1 \phi(u,0)\, u\, du}{(1+\beta)^2} H_0 \tag{21}$$

where $\Gamma \equiv R_{out}/d$, $\beta \equiv \frac{R_{in}}{R_{out}}$ and $\eta \approx 0.65$

Numerical Simulation

Matrix Collocation Method

To solve the mixed boundary value problem without analytical approximations, we discretize the continuous Hankel transform into a linear system. We evaluate the domain at N discrete radial points r_i and wavenumbers k_j.

Constructing the System

The three integral equations form the rows of our matrix $\mathrm{M}\vec{A} = \vec{b}$:

- **Inner ($r_i < \beta$):** $\mathrm{M}_{i,j} = \coth(k_j d) J_0(k_j r_i) \Delta k = 0$
- **Ring ($\beta \leq r_i \leq 1$):** $\mathrm{M}_{i,j} = k_j J_0(k_j r_i) \Delta k = -1$
- **Outer ($r_i > 1$):** $\mathrm{M}_{i,j} = \coth(k_j d) J_0(k_j r_i) \Delta k = 0$

Because Fredholm equations of the first kind are naturally ill-conditioned, we use Least Squares (`lstsq`) rather than direct inversion to find the spectral amplitude $A(k)$.

Solved using **NumPy**, **SciPy** (j_0 function) and plotted using **MatPlotLib**

Numerical Simulation

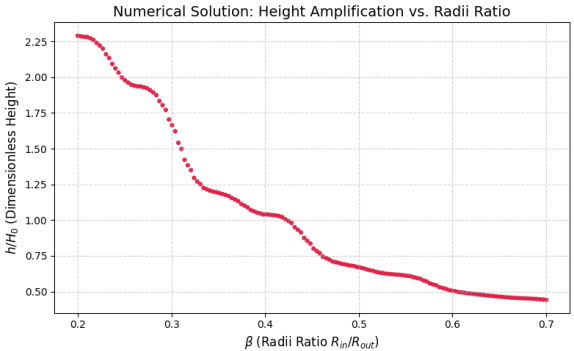


Figure: Simulation Result for a Given $\Gamma = 2$

- The function is **decreasing** with β as predicted, and has local maxima
- The goal of our experiment will be to find for what beta is the theoretical trend relevant

Experimental Setup

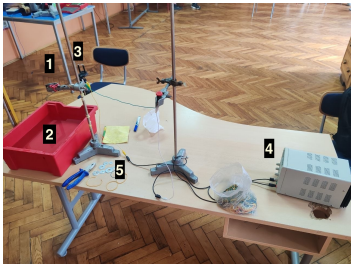


Figure: Experimental Setup

- **1:** Electromagnetic release with height change for verticality and zero rotation.
- **2:** $40 \times 30 \times 12$ cm acrylic tank.
- **3:** Stand with phone with 240 fps camera and a ruler.
- **4:** Current source
- **5:** Set of rings with different β.

Experimental Setup: Ring Geometries

Selection of Rings

We combined different DIN standards.

DIN Standard	**Type**	d_{in} **[mm]**	D_{out} **[mm]**	h **[mm]**	**Mass** m **[g]**	β
DIN 9021	Wide	10.5	30.0	2.5	12.2	**0.350**
DIN 7349	Thick	8.4	21.0	4.0	9.1	**0.400**
DIN 125A	Standard	10.5	20.0	2.0	3.6	**0.525**
DIN 433	Narrow	13.0	20.0	2.0	2.8	**0.650**
DIN 433	Narrow	17.0	24.0	2.0	3.5	**0.708**

Table: Physical dimensions and calculated radii ratio $\beta = R_{in}/R_{out}$.

Experimental Setup: Measurements

Maximum Height

As the maximum height we'll define the **average height** of the jet's subsequent motion (not counting in the pitch off droplets), which is measured using Tracker (manually), and the we **multiply it by 2**.

Measurements

Measurements were taken only when the ring flew down with no angular deflection. For every set of parameters 5 measurements were taken and averaged upon, and plotted error bars are taken as maximal deviations from the averages.

Results: h vs H_0

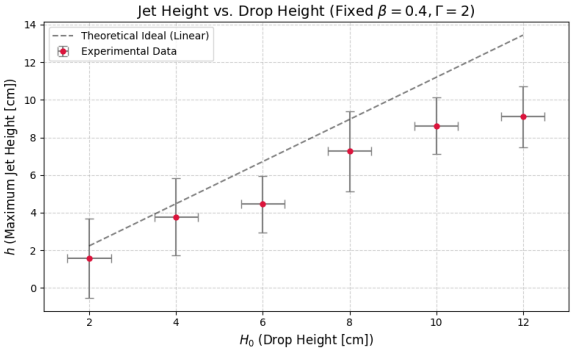


Figure: Graph of h vs H_0

Results: Height Amplification (h/H_0) vs Radii Ratios (β)

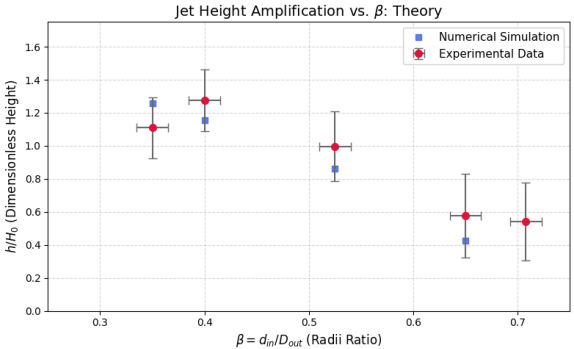


Figure: Graph of Height Amplification vs β for $\Gamma = 2, H_0 = 8\text{cm}$

Results: Depth (Γ) Influence on Height Amplification (h/h_0)

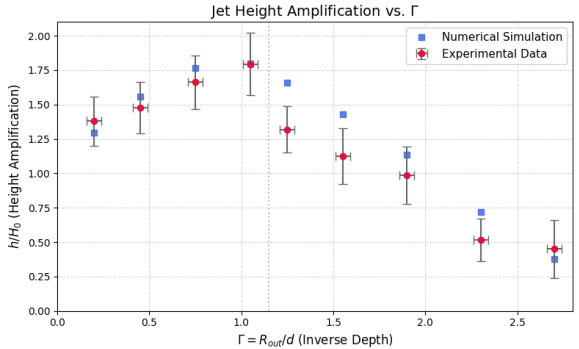


Figure: Graph of Heigth Amplification vs Γ for $\beta = 0.4$, $H_0 = 8\text{cm}$

Discussion

- Theory as an upper bound.
- Dissipation factors
- Imperfect apparatus requires many trials
- Theory predicts the trends that experiment follows

Appendix: Properties of the Radial Solution

Selection of $J_0(kr)$

The radial component of Laplace's equation yields a general solution of the form $C_1 J_n(kr) + C_2 Y_n(kr)$. We restrict this solution based on physical constraints:

- **Finiteness at Origin:** $Y_n(kr) \to -\infty$ as $r \to 0$. To maintain finite velocity potential at the center axis, $C_2 = 0$.
- **Axisymmetry:** The vertical drop ensures the potential is independent of the azimuthal angle θ, requiring the order $n = 0$.

Thus, the radial basis function is strictly $J_0(kr)$.

The Role of the Hankel Transform

A single Bessel function cannot satisfy the discontinuous mixed boundary conditions at $z = 0$ (the sharp edges of the ring). The integral over $k \in [0, \infty)$ acts as a continuous superposition of radial spatial frequencies, allowing the reconstruction of the localized impact zone.

Appendix: The Zeroth-Order Hankel Transform

Definition and Necessity

The Hankel transform is an integral transform whose kernel is a Bessel function. For systems with cylindrical symmetry ($\frac{\partial}{\partial \theta} = 0$), the zeroth-order Hankel transform is the natural choice to resolve the radial dependence:

$$H_0[f(r)] = \tilde{f}(k) = \int_0^\infty f(r) J_0(kr) r \, dr \tag{22}$$

with the corresponding inverse transform:

$$f(r) = \int_0^\infty \tilde{f}(k) J_0(kr) k \, dk \tag{23}$$

Physical Interpretation

Just as a Fourier transform decomposes a signal into sines and cosines, the Hankel transform decomposes the fluid's potential field into a **superposition of cylindrical waves** (J_0) with radial wavenumbers k.

Appendix: Algebraic Reduction of the Laplacian

Transforming the PDE into an ODE

The power of the Hankel transform lies in its operational property. It transforms the radial part of the Laplacian operator into a simple algebraic term:

$$H_0\left[\frac{\partial^2\phi}{\partial r^2}+\frac{1}{r}\frac{\partial\phi}{\partial r}\right]=-k^2\tilde{\phi}(k,z) \tag{24}$$

Substituting this into the 3D Laplace equation ($\nabla^2\phi=0$) reduces the partial differential equation to a linear second-order ordinary differential equation (ODE) in the z-direction:

$$\frac{d^2\tilde{\phi}}{dz^2}-k^2\tilde{\phi}=0 \tag{25}$$

Conclusion: This transformation allows us to solve for the vertical structure of the flow ($Z(z)$) analytically, leaving only the radial "spectral" amplitude $A(k)$ to be determined.

Appendix: Mapping to the Spectral Domain

Handling Discontinuities

The "complex" boundary conditions at the surface (sharp edges of the ring) are difficult to solve in physical space (r). However, in the **spectral domain** (k), the Hankel transform allows us to represent these discontinuities as a system of integral equations:

- The spatial "Top-Hat" velocity profile under the ring is mapped to a weighted sum of Bessel functions.
- The amplitude function $A(k)$ acts as a weight, determining how much "information" from each radial frequency k is required to satisfy the ring's geometry and the tank's depth simultaneously.

Appendix: Numerical Simulation 1

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import j0

N = 1000
k = np.linspace(0.05, 40, N)
dk = k[1] - k[0]
r = np.linspace(0.01, 2.5, N)

d_R = 1.5
Gamma = 1 / d_R
eta = 0.65

betas = np.linspace(0.2, 0.7, 150)
h_ratios = []

for beta in betas:
    M = np.zeros((N, N))
    b = np.zeros(N)

    for i in range(N):
        if r[i] < beta or r[i] > 1.0:
            M[i, :] = j0(k * r[i]) * dk
        else:
```

Appendix: Numerical Simulation 2

Python Code

```
            M[i, :] = k * np.tanh(k * d_R) * j0(k * r[i]) * dk
            b[i] = -1.0

    X, _, _, _ = np.linalg.lstsq(M, b, rcond=None)

    mask = (r >= beta) & (r <= 1.0)
    r_ring = r[mask]
    phi_ring = np.array([np.sum(X * j0(k * r_val) * dk) for r_val in r_ring])

    integral_val = np.trapz(np.abs(phi_ring) * r_ring, r_ring)

    h_H0 = (8 * eta * Gamma * integral_val) / (1 + beta)**2
    h_ratios.append(h_H0)

plt.figure(figsize=(8, 5))
plt.scatter(betas, h_ratios, color='crimson', s=20, alpha=0.9, edgecolor='none')
plt.title('Numerical Solution: Height Amplification vs. Radii Ratio', fontsize=14)
plt.xlabel(r'$\beta$ (Radii Ratio $R_{in}/R_{out}$)', fontsize=12)
plt.ylabel(r'$h/H_0$ (Dimensionless Height)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

Appendix: Green's First Identity

1. The General Theorem

Green's First Identity relates the volume integral of a field to the surface integral bounding that volume. For any two scalar functions ϕ and ψ:

$$\iiint_V \left(\phi \nabla^2 \psi + \nabla\phi \cdot \nabla\psi\right) dV = \iint_S \phi \frac{\partial \psi}{\partial n} dS \tag{26}$$

where V is the fluid volume, S is its bounding surface, and n is the outward-pointing normal vector.

2. Application to Potential Flow

In our irrotational, incompressible model, we set $\psi = \phi$. Because the flow satisfies Laplace's equation ($\nabla^2 \phi = 0$), the first term vanishes perfectly:

$$\iiint_V |\nabla\phi|^2 dV = \iint_S \phi \frac{\partial \phi}{\partial n} dS \tag{27}$$

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Most Important Slide



Figure: +2 Points for One Who Guesses His Name